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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz,		

Task Description:	Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility as well as to obtain indicators in the specific heat of a cooperative cross over behavior in the shear viscosity at high temperatures. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation will be evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements and with predictions from Kelton's coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL, Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds will also be made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature using the MSL-EML on the ISS. These will be coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalescece measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which has the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant will allow these data to be obtained.
	Following is a brief summary of the activities carried out under NASA grant NNX16B52G over the past year. Experiments on the International Space Station (ISS) – Experiments on two batch 1 samples, Vit 106 (Zr57Nb5Al10Cu15.4Ni12.6, ID#21) were conducted during December 2015 to January 2016 and Vit106a (Zr58.5Nb2.8Cu15.6Ni12.8Al10.3, ID#13) in May 2016. Nucleation, specific heat, viscosity, and surface tension for the equilibrium and supercooled liquids were studied. The Vit106 sample showed better performance compared to the Vit106a. Since Vit106a was one of the very first samples processed with the MSL-EML, it is now suspected that it was contaminated by some residual oxygen in the lines connecting the gas tanks to the processing chamber; this might have caused lower supercooling (110 degrees C) compared to Vit106 (235 degrees C) and a large scatter in the viscosity data. The major findings for the Vit106 sample are: a) large supercoolings consistent with expectations; b) a specific heat anomaly above the liquidus at temperatures close to where a crossover behavior was observed in the viscosity in ground-based studies; c) viscosity data consistent with the ground-based results. More studies are planned in vacuum and gas atmospheres as a further check of the specific heat anomaly and of nucleation as a function of stirring. A request has been made to the Internal Working Group (IWG) for additional coil-deposition for these studies. Batch 2 Flight Samples for the ISS Experiments – Ti39.5Zr39.5Ni21 (sample ID # 39), a quasicrystal-forming alloy, is a designated flight sample for the batch 2 experiments. Three 6.0 mm diameter samples of this composition were made at Washington University and delivered to Airbus in November 2015 for shipment to the ISS. Similarly, three 6.5 mm

Task Progress:	diameter bulk metallic glass-forming Zr47Cu47Al6 samples, which is a back-up alloy for batch 2 studies, were delivered to Airbus in December 2015. Parabolic flight samples - Based on the results of the nucleation studies made on the ISS during December-January 2015-2016, it was determined that resistivity data for liquid Vit106 (Zr57Nb5Al10Cu15.4Ni12.6) were needed to estimate the stirring in the liquid. Such measurements are planned in parabolic flight experiments and the required samples were sent to DLR in July 2016. Preparation for experiments on batch 2 samples – Trips were made by Anup Gangopadhyay during January 2016 to Koln for the flight experiments parameter checks for Ti39.5Zr39.5Ni21 (ID #39) and Cu50Zr50 (ID #27). These were conducted in collaboration with the members of the ICOPROSOL, PARSEC, THERMOLAB, and ELFSTONE for the first sample and with MULTIPHAS for the second. Other activities with ESA linked to the ISS experiments – The Principal Investigator (PI) attended the IWG 13 meeting at the DLR in Cologne, Germany, during December 15-16 to discuss the results from Vit106 experiments. Concerns were raised by several members of gas-contamination during this experiment and methods to test this hypothesis were discussed. The PI also attended the 14th IWG meeting in Koln during May 24-25. The main agenda items were the results of testing on the operating and flight modules of the MSL-EML conducted by MUSC and Airbus. Final preparation for the batch 2 experiments was also discussed. Ground-based activities in support of the ISS experiments – Ground-based studies on many different properties of the equilibrium and supercooled liquids were conducted. To gain a broader perspective, many other related glass-forming alloys were also investigated. The major highlights are briefly described below: 1. Liquid fragility from thermal expansion measurements – The liquid fragility is important for glass formation and nucleation and growth of crystals from the liquid. Liquids are characterized as strong or fragile, depending on how fast their dynamical properties (viscosity, structural relaxation time) evolve with temperature near the glass transition temperature, Tg. Rarely, far above Tg are these considered for characterization in the equilibrium and supercooled liquids. From thermal expansion measurements of a large variety of marginal and good glass-forming liquids, we have established that the liquid fragility is closely related to that part of the thermal expansion coefficient that is caused by structural changes with temperature. This is a significant new result. 2. Liquid fragility and structure parameters – The connection between liquid fragility and structure has been a central interest of this research for some time. It was established in our earlier investigation that the rapid change in liquid structure near Tg is intimately connected to the kinetic fragility. It is now established that the rates of change with temperatures of the liquid structure factor, S(q), and the pair correlation function, g(r), in equilibrium and supercooled liquids are determined by the liquid fragilities. A clear connection between structure and dynamical properties, which has been questioned in some recent theoretical studies, is clearly established experimentally. 3. What determines liquid fragility? – There is little consensus on what, at the fundamental level, determines liquid fragility. We have found that to a large extent fragility is determined by the cohesive energy, providing a useful method for estimating liquid fragility without measuring it. In a parallel study, we have found that the repulsive part of the atomic interaction potential also correlates with fragility. Taken together, these experimental results provide comprehensive evidence that the fragility is related to the nature of the atomic interaction. Our studies also demonstrate that strong liquids are not necessarily good glass formers, which is in disagreement with the prevalent idea among a large section of the metallic glass community. 4. Ongoing experiments on liquid structure – A three week X-ray scattering experiment was carried in June 2016 at the advanced photon source at the Argonne National Laboratory using the BESL technique. A large number of marginal and good glass-forming alloys were examined, including those selected for future ISS experiments. The analysis of this data is in progress to determine the structures of these liquids and of the corresponding glasses. In parallel, neutron scattering experiments have been performed on a subset of these alloys at the Oak-Ridge National laboratory using the NESL technique. These data are also being analyzed. By combining the neutron and X-ray results, we will be able to investigate the chemical and topological ordering of the liquids planned for future ISS experimental study. This will be important in correlating changes measured in the specific heat and viscosity with structural/chemical ordering.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
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